MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY TEST – 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 4 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Short Answer Type-I question of 2 marks.
4. **Section C** comprises 3 Short Answer Type-II questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (5 × 1 = 5 Marks)

Q1. The value of $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ is equal to: [1]

(A) $\cos 60^\circ$ (B) $\sin 60^\circ$
(C) $\tan 60^\circ$ (D) $\sin 30^\circ$

Q2. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = \frac{1}{\sqrt{3}}$, where $0^\circ < A+B \leq 90^\circ$ and $A > B$, then the value of A is: [1]

(A) 45° (B) 60° (C) 30° (D) 15°

Q3. In the previous question, the value of acute angle B is: [1]

(A) 15° (B) 30° (C) 45° (D) 60°

Q4. The value of $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$ is: [1]

(A) 1 (B) 2 (C) 0 (D) $\sqrt{3}$

Q5. **Assertion (A):** For any acute angle θ, $\sin^2\theta + \cos^2\theta = 1$.
Reason (R): $\sec^2\theta - \tan^2\theta = 1$ for all acute angles $0^\circ \leq \theta < 90^\circ$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type-I (1 × 2 = 2 Marks)

Q6. If $\tan(A+B) = \sqrt{3}$ and $\tan(A-B) = 1$, where $0^\circ < A+B \leq 90^\circ$ and $A > B$, find the values of A and B. [2]

SECTION C: Short Answer Type-II (3 × 3 = 9 Marks)

Q7. Prove that: $\frac{1+\cos\theta+\sin\theta}{1+\cos\theta-\sin\theta} = \frac{1+\sin\theta}{\cos\theta}$. [3]

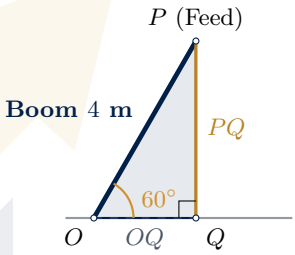
Q8. Prove that: $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta} = 2\csc\theta$. [3]

Q9. Prove that: $(\sec A - \tan A)^2 = \frac{1-\sin A}{1+\sin A}$. [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Satellite Earth Station Parabolic Antenna Alignment

A telecommunications ground station uses an adjustable parabolic dish antenna. A rigid calibration boom OP of length 4 m extends from the antenna vertex O to feed horn P. When tracking a geostationary satellite, the boom is inclined at an elevation angle of $\theta = 60^\circ$ to the horizontal base line OQ.



Based on the above geometric model, answer the following questions:

(i) Calculate the vertical height PQ of the feed horn above the horizontal base. [1]

(ii) Calculate the horizontal distance OQ from the dish vertex to the base of the feed horn projection. [1]

(iii) Verify that $OP^2 = OQ^2 + PQ^2$ using the values obtained, and express $\tan 60^\circ$ as the ratio $\frac{PQ}{OQ}$. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★